

SOP #12 – ATI (Analysis to Improve) for Exhaust Fan Systems

Goal: Increase exhaust effectiveness, reduce energy waste, extend equipment life

1. Performance Benchmarks

Metric	Ideal Value/Range	Optimization Strategy
Fan Runtime Efficiency	>85% of design airflow	Tune motor speed and damper timing
Power Consumption	Minimized per CFM	Use VFDs, adjust for occupancy loads
Preventive Maintenance Score	100% completion	Monthly log and review schedule
Static Pressure vs. Design	Within 10% of target	Check filters, duct layout, leakage

2. Improvement Measures

Install Variable Frequency Drives (VFDs): Improve control, reduce kWh usage
 Duct Sealing and Layout Correction: Prevent pressure losses
 Sound/Vibration Dampers: Extend motor and bearing life
 Air Quality Sensors: Control fan runtime based on VOC or CO₂ levels
 Digital Fan Runtime Monitors: Automate diagnostics and maintenance alerts

3. Compliance & Safety Notes

ASHRAE 62.1: Minimum ventilation rate standards
 OSHA 1910 Subpart D: Covers access, guards, and fall hazards
 NFPA 91: Exhaust system design and fire prevention
 NEC Wiring Requirements: Electrical circuit rating and grounding
 Maintenance Logs: Must be retained for inspections and safety audits